

NLP Final Report

November 2025

Abstract

This report presents our approach to the problem of predicting attribute values from product text in e-commerce catalogs. We focus on the IndoML Datathon dataset, which contains both product titles and receipt-style descriptions. Our initial plan was to begin with simple NLP baseline models and then progress to transformer-based and LLM-based models later.

1 Introduction

E-commerce platforms depend on structured product attributes such as category and brand to organize catalogs and power search and recommendations. However, product listings often contain only short and noisy text, and many sellers avoid filling in all attributes due to the manual effort required. This results in incomplete product listings, making it harder for platforms to maintain their catalogs effectively.

In this project, we address this issue by predicting missing attributes from product text:

- **Attributes:** Structured labels such as hierarchical categories (L0–L4 or Supergroup $\rightarrow$ Group $\rightarrow$ Module) and brand.
- **Text/Description:** Unstructured input such as product titles (*“Enclume Angled Pot Hook, Set of 6”*) or receipt-style descriptions (*“1 car mat set”*).

2 Approaches

2.1 Hybrid Global–Local Supervision for Hierarchical Product Classification

To address the hierarchical classification challenge, we adopted a hybrid “global + local” methodology, drawing inspiration from the core philosophy presented in Wehrmann et al. (2018). The essential idea advocated by their work is to design a model “capable of simultaneously optimizing both local and global loss functions” (Wehrmann et al., 2018, p. 1). While Wehrmann et al. propose a

specific complex architecture (HMCN), our `GECEClassifier` implements this hybrid-loss philosophy using a more streamlined and robust multi-task architecture.

Our model’s architecture begins with a shared 768-dimensional DistilBERT embedding for each product. This embedding is passed through a shared MLP backbone, which generates a 256-dimensional learned representation. This shared representation is then fed in parallel to two distinct types of classification heads:

1. **Local Heads:** A `ModuleDict` of independent linear layers, one for each individual attribute (Brand, L0, L1, L2, L3, and L4). These heads are trained to optimize the prediction for each specific, fine-grained attribute.
2. **Global Head:** A single linear layer that predicts the `global_path`, which is the complete, concatenated hierarchical path (e.g., "L0 — L1 — ... — L4"). This head is trained to understand the full dependency structure of the hierarchy.

The model’s training is guided by a composite loss function, which is the simple sum of all individual losses. This aligns directly with the "hybrid" concept from Wehrmann et al. (2018):

$$\mathcal{L}_{total} = \mathcal{L}_G + \mathcal{L}_L$$

Here, $\mathcal{L}_G$ is the "global loss" (the Cross-Entropy Loss from the `global_head`), and $\mathcal{L}_L$ is the "local loss" (the *sum* of the Cross-Entropy Losses from all individual `local_heads`). This hybrid-loss approach allows the model to learn the "global information" of the entire hierarchy from $\mathcal{L}_G$ while simultaneously reinforcing the "local hierarchical class-relationships" at each individual level via $\mathcal{L}_L$.

Training Efficiency: The GECE Classifier demonstrated excellent training efficiency, completing 30 epochs in approximately 20 minutes using pre-computed DistilBERT embeddings. This rapid training cycle enabled extensive experimentation and hyperparameter tuning.

2.2 Hierarchical BERT with Constraint Masking

2.2.1 Overview and Concept

The Hierarchical BERT Classifier is designed for multi-level classification where predictions follow nested category structures. Instead of treating each category independently, the model leverages parent-child relationships between labels (L0→L1→L2→L3→L4→Brand) to enhance prediction accuracy and ensure logical consistency. This approach reflects real-world e-commerce categorization where categories form natural hierarchies, preventing impossible predictions and reducing label noise.

2.2.2 Architecture

We developed a hierarchical BERT-based classifier that enforces logical constraints between category levels. The architecture uses BERT-tiny as the backbone feature extractor, followed by parallel classification heads for each attribute level (L0-L4, brand). The key innovation is **hierarchical masking** - during inference, the model uses parent category predictions to constrain possible child categories through learned dependency maps.

The architecture consists of three main components:

- **Feature Extraction:** BERT-tiny transformer backbone processes input text and extracts contextual representations using the [CLS] token embedding
- **Multi-Head Classification:** Separate linear layers (heads) for each hierarchy level: L0, L1, L2, L3, L4, and Brand, each outputting logits for its possible classes
- **Hierarchical Masking Logic:** Masking dictionaries that specify valid class indices for every parent label, enforcing valid parent-child relationships

2.2.3 Data and Preprocessing

- **Input:** Product titles concatenated with store information
- **Tokenization:** BERT tokenizer with max length 128 tokens
- **Label Encoding:** SKLearn LabelEncoder for all categorical attributes
- **Hierarchical Mapping:** Learned parent-child relationships from training data (e.g., L0→L1, L1→L2 mappings)
- **Store-Brand Constraints:** Store-specific brand availability mappings

2.2.4 Working Methodology

The model operates in two distinct modes:

Training Phase:

- Forward pass through BERT backbone to obtain [CLS] token embeddings
- Parallel prediction of all attribute levels using **true labels** for constraint masking
- Multi-task loss optimization: $\mathcal{L} = \mathcal{L}_{L0} + \mathcal{L}_{L1} + \mathcal{L}_{L2} + \mathcal{L}_{L3} + \mathcal{L}_{L4} + \mathcal{L}_{brand}$
- All hierarchy heads trained jointly with cross-entropy losses computed for each head

Inference Phase:

- Predict L0 category first, then use prediction to mask invalid L1 options
- Iteratively apply hierarchical constraints: $L0 \rightarrow L1 \rightarrow L2 \rightarrow L3 \rightarrow L4$ using **pre-dicted labels** for masking
- Apply store-based brand constraints when available
- Ensures logically consistent category trees by avoiding impossible combinations

2.2.5 Parameters and Implementation

- **Model:** BERT-tiny (4 layers, 128 hidden size, 2 attention heads)
- **Batch Size:** 128
- **Learning Rate:** $1e-5$
- **Epochs:** 10
- **Training Time:** Approximately 1.5 hours per epoch, totaling 15 hours for full training
- **Optimizer:** Adam with mixed-precision training using GradScaler
- **Loss Function:** CrossEntropyLoss for each hierarchy level
- **Hardware:** 2 Kaggle T4 GPUs via DataParallel
- **Category Counts:** L0: 10, L1: 67, L2: 292, L3: 944, L4: 2173, Brands: 4656
- **Constraint Maps:** 10 $L0 \rightarrow L1$ mappings, 67 $L1 \rightarrow L2$ mappings, 292 $L2 \rightarrow L3$ mappings, 944 $L3 \rightarrow L4$ mappings, and store-brand mappings

2.2.6 Advantages of Hierarchical Approach

- **Reflects Real-World Structure:** E-commerce categories naturally form hierarchies that this model explicitly captures
- **Improves Accuracy:** By focusing only on relevant subcategories and avoiding impossible predictions
- **Ensures Consistency:** Maintains logical flow between category levels, reducing prediction noise
- **Efficient Training:** Joint optimization of all hierarchy levels with constraint-aware masking

This approach ensures that predictions respect the inherent hierarchical structure of e-commerce categories while leveraging BERT’s powerful language understanding capabilities, resulting in more accurate and logically consistent attribute predictions.

2.3 Multi-Task Flat Classification with Hierarchy-Aware Cascaded Inference

To model the product taxonomy, we construct a multi-task classifier operating on 768-dimensional DistilBERT embeddings. A lightweight MLP maps each embedding into a shared 256-dimensional representation, which is then fed into six parallel classification heads corresponding to `details.Brand` and the hierarchical levels L0–L4. Each head is trained independently using cross-entropy loss, following the well-established "local classifier per node/level" paradigm widely discussed in hierarchical classification literature (Silla and Freitas, 2011). Importantly, we do *not* impose hierarchical constraints during training: several studies (Cesa-Bianchi et al., 2006) have shown that training-time hierarchical enforcement can propagate early-level errors, cause optimization instability, and reduce generalization. Instead, we follow the standard practice of applying hierarchy only at inference. After predicting L0, we apply a masked decoding procedure in which the candidate set for each deeper level (L1–L4) is restricted to children observed under the predicted parent in the training taxonomy. This cascaded masking enforces consistency with the hierarchy while preserving the robustness of independent multi-task training. Brand prediction remains independent, as it is not part of the category tree. This approach yields a stable and computationally efficient model that aligns with best practices in hierarchical classification.

Training Efficiency: The Hierarchical Masked Inference approach demonstrated excellent computational efficiency, with the complete training process (30 epochs) requiring only approximately 20 minutes. This efficiency stems from the use of precomputed DistilBERT embeddings, which eliminates the need for expensive forward passes through the transformer backbone during classifier training.

3 Results and Evaluation

3.1 Performance Metrics Overview

We evaluated three different approaches on the validation dataset: Hierarchical BERT with Constraint Masking, GECE Classifier, and Hierarchical Masked Inference. All models demonstrate strong performance across hierarchical levels (L0–L4) and brand prediction, with each approach showing distinct strengths and characteristics.

3.2 Hierarchical BERT with Constraint Masking Performance

3.2.1 L0 Category Performance

- **Accuracy:** 97.16% - Excellent overall classification performance
- **Macro Precision:** 69.82% - Good performance across all L0 classes

- **Macro Recall:** 65.39% - Consistent recall across different L0 categories
- **Macro F1-Score:** 66.92% - Balanced precision and recall

3.2.2 L1 Category Performance

- **Accuracy:** 94.81% - Strong performance with more fine-grained categories
- **Macro Precision:** 86.00% - High precision in L1 classification
- **Macro Recall:** 76.21% - Good recall across 67 L1 categories
- **Macro F1-Score:** 79.56% - Well-balanced performance

3.2.3 L2 Category Performance

- **Accuracy:** 92.30% - Maintains strong performance with 292 categories
- **Macro Precision:** 80.74% - Good precision despite increased complexity
- **Macro Recall:** 71.02% - Consistent recall across many categories
- **Macro F1-Score:** 73.75% - Maintains good balance

3.2.4 L3 Category Performance

- **Accuracy:** 88.82% - Good performance with 944 fine-grained categories
- **Macro Precision:** 76.24% - Respectable precision at this granularity
- **Macro Recall:** 68.18% - Adequate recall for highly specific categories
- **Macro F1-Score:** 69.94% - Reasonable performance given complexity

3.2.5 L4 Category Performance

- **Accuracy:** 91.57% - Strong recovery in accuracy for most specific level
- **Macro Precision:** 75.25% - Good precision across 2173 categories
- **Macro Recall:** 68.55% - Consistent recall pattern
- **Macro F1-Score:** 69.61% - Maintains performance balance

3.2.6 Brand Prediction Performance

- **Accuracy:** 98.62% - Outstanding brand classification performance
- **Macro Precision:** 98.45% - Near-perfect precision across 4656 brands
- **Macro Recall:** 98.11% - Excellent recall for brand identification
- **Macro F1-Score:** 98.15% - Exceptional overall brand prediction

3.3 GECE Classifier Performance

3.3.1 Brand Prediction Performance

- **Accuracy:** 93.78% - Strong brand classification performance
- **Precision:** 94.16% - High precision in brand identification
- **Recall:** 93.78% - Consistent recall across brands
- **F1-score:** 93.54% - Well-balanced brand prediction

3.3.2 L0 Category Performance

- **Accuracy:** 91.24% - Excellent top-level category performance
- **Precision:** 91.21% - High precision for broad categories
- **Recall:** 91.24% - Consistent recall across L0 categories
- **F1-score:** 91.16% - Strong overall performance

3.3.3 L1 Category Performance

- **Accuracy:** 89.16% - Strong performance with finer granularity
- **Precision:** 89.12% - Maintains high precision
- **Recall:** 89.16% - Good recall across subcategories
- **F1-score:** 89.00% - Balanced performance

3.3.4 L2 Category Performance

- **Accuracy:** 88.81% - Consistent performance at deeper hierarchy
- **Precision:** 88.92% - Slightly higher precision than accuracy
- **Recall:** 88.81% - Stable recall performance
- **F1-score:** 88.64% - Maintains good balance

3.3.5 L3 Category Performance

- **Accuracy:** 86.86% - Good performance with high granularity
- **Precision:** 87.20% - Strong precision for specific categories
- **Recall:** 86.86% - Adequate recall performance
- **F1-score:** 86.66% - Respectable overall performance

3.3.6 L4 Category Performance

- **Accuracy:** 90.06% - Excellent performance at most specific level
- **Precision:** 90.24% - High precision for fine-grained categories
- **Recall:** 90.06% - Strong recall performance
- **F1-score:** 89.80% - Well-balanced metrics

3.3.7 Global Path Performance

- **Accuracy:** 87.06% - Strong overall hierarchical path prediction
- **Precision:** 87.48% - Good precision for complete category paths
- **Recall:** 87.06% - Consistent recall performance
- **F1-score:** 86.76% - Balanced global prediction

3.4 Hierarchical Masked Inference Performance

3.4.1 Brand Prediction Performance

- **Accuracy:** 95.50% - Excellent brand classification with constraint enforcement
- **Precision:** 95.71% - High precision in brand identification
- **Recall:** 95.50% - Strong recall performance
- **F1-score:** 95.33% - Well-balanced brand prediction
- **R² Score:** 0.9011 - Strong explanatory power

3.4.2 L0 Category Performance

- **Accuracy:** 92.04% - Strong top-level category performance
- **Precision:** 92.00% - High precision for broad categories
- **Recall:** 92.04% - Consistent recall across L0 categories
- **F1-score:** 91.96% - Excellent overall performance
- **R² Score:** 0.7963 - Good explanatory power

3.4.3 L1 Category Performance

- **Accuracy:** 87.06% - Solid performance with finer granularity
- **Precision:** 87.01% - Maintains high precision
- **Recall:** 87.06% - Good recall across subcategories
- **F1-score:** 86.79% - Balanced performance
- **R² Score:** 0.7351 - Reasonable explanatory power

3.4.4 L2 Category Performance

- **Accuracy:** 83.44% - Consistent performance at deeper hierarchy
- **Precision:** 83.31% - Slightly lower precision than accuracy
- **Recall:** 83.44% - Stable recall performance
- **F1-score:** 82.89% - Maintains good balance
- **R² Score:** 0.6738 - Moderate explanatory power

3.4.5 L3 Category Performance

- **Accuracy:** 79.93% - Good performance with high granularity
- **Precision:** 79.92% - Strong precision for specific categories
- **Recall:** 79.93% - Adequate recall performance
- **F1-score:** 79.16% - Respectable overall performance
- **R² Score:** 0.5959 - Moderate explanatory power

3.4.6 L4 Category Performance

- **Accuracy:** 84.92% - Strong recovery at most specific level
- **Precision:** 84.91% - High precision for fine-grained categories
- **Recall:** 84.92% - Excellent recall performance
- **F1-score:** 84.30% - Well-balanced metrics
- **R² Score:** 0.6108 - Moderate explanatory power

Level	Hierarchical BERT			GECE Classifier			Masked Inference	
	Acc	Prec	F1	Acc	Prec	F1	Acc	F1
Brand	98.62%	98.45%	98.15%	93.78%	94.16%	93.54%	95.50%	95.33%
L0	97.16%	69.82%	66.92%	91.24%	91.21%	91.16%	92.04%	91.96%
L1	94.81%	86.00%	79.56%	89.16%	89.12%	89.00%	87.06%	86.79%
L2	92.30%	80.74%	73.75%	88.81%	88.92%	88.64%	83.44%	82.89%
L3	88.82%	76.24%	69.94%	86.86%	87.20%	86.66%	79.93%	79.16%
L4	91.57%	75.25%	69.61%	90.06%	90.24%	89.80%	84.92%	84.30%

Table 1: Comparative Performance Metrics Across All Approaches

Approach	Epochs	Time per Epoch	Total Time	Training Strategy
Hierarchical BERT	10	1.5 hours	15 hours	End-to-end BERT fine-tuning
GECE Classifier	30	0.67 minutes	20 minutes	Frozen embeddings + classifier
Masked Inference	30	0.67 minutes	20 minutes	Frozen embeddings + cascaded inference

Table 2: Training Efficiency Comparison Across Approaches

3.5 Comparative Performance Analysis

3.6 Training Efficiency Comparison

3.7 Key Observations and Analysis

3.7.1 Performance Trends Across Models

- **Hierarchical BERT Superiority:** Demonstrates highest accuracy across all levels (88.82-98.62%), particularly excelling in brand prediction, but requires significantly more training time (15 hours total)
- **GECE Classifier Consistency:** Shows balanced performance across hierarchy levels with strong precision-recall metrics, achieving good results with only 20 minutes of training
- **Masked Inference Trade-offs:** Provides good performance with explicit constraint enforcement but shows performance degradation in middle hierarchy levels, with efficient 20-minute training
- **Brand Prediction Excellence:** All models achieve strong brand prediction (>93%), with Hierarchical BERT reaching exceptional 98.62% accuracy

3.7.2 Hierarchical Performance Patterns

- **Top-Level Dominance:** All models perform best at L0 and brand levels, demonstrating effective broad categorization
- **Mid-Level Challenge:** Performance typically decreases at L2-L3 levels where category granularity increases significantly

- **Bottom-Level Recovery:** L4 shows performance recovery in most models, suggesting effective handling of fine-grained categories
- **Constraint Effectiveness:** Models with explicit hierarchical constraints show more consistent performance across levels

3.7.3 Architectural Strengths

- **Hierarchical BERT:** Excels through end-to-end training and explicit constraint enforcement during inference, but requires substantial computational resources
- **GECE Classifier:** Demonstrates robust performance with global-local architecture and balanced metrics, offering excellent training efficiency
- **Masked Inference:** Provides good performance with modular design and post-hoc constraint application, enabling rapid experimentation
- **Training Efficiency:** GECE and Masked Inference approaches offer 45x faster training through precomputed embeddings compared to end-to-end BERT fine-tuning

3.8 Conclusion on Model Effectiveness

All three hierarchical classification approaches demonstrate strong performance for e-commerce attribute prediction, each with distinct advantages and trade-offs:

- **Hierarchical BERT Advantages:** Superior overall accuracy (88.82-98.62%), exceptional brand prediction, and integrated constraint enforcement through end-to-end training, making it ideal for maximum accuracy scenarios where training time is not a constraint
- **GECE Classifier Advantages:** Balanced precision-recall metrics, global path understanding, and exceptional training efficiency (20 minutes for 30 epochs), making it suitable for rapid prototyping and scenarios requiring quick iteration
- **Masked Inference Advantages:** Modular design, explicit constraint application, strong brand prediction (95.50%) with good overall performance, and the same efficient 20-minute training as GECE, providing a practical balance of performance and efficiency
- **Practical Utility:** All models achieve accuracy rates above 79% across all levels, making them suitable for real-world e-commerce applications with different resource constraints
- **Architecture Validation:** The results validate multiple effective strategies for hierarchical classification, from computationally intensive end-to-end training to highly efficient modular approaches

The comparative analysis reveals a clear trade-off between performance and training efficiency. While end-to-end Hierarchical BERT achieves the highest absolute performance, the modular approaches (GECE and Masked Inference) offer compelling 45x improvements in training time with only moderate performance reductions. This diversity of effective approaches provides flexibility for different deployment scenarios, from research environments requiring rapid experimentation to production systems where maximum accuracy is paramount.